

Controllability :

A state x is said to be controllable if, given any $x_0 \neq 0$ and any finite time interval $[0, T]$, there exists an input $\{u(t), t \in [0, T]\}$ such that $x(T) = 0$. If all states are controllable, then the system is said to be completely controllable.

Reachability.

A state $\bar{x} \neq 0$ is said to be reachable (from origin) if, given $x(0) = 0$, there exists a finite time interval $[0, T]$ and an input $\{u(t), t \in [0, T]\}$, such that $x(T) = \bar{x}$. If all states are reachable, the system is said to be completely reachable.

Test for controllability

$$\text{for } \dot{x} = Ax + Bu$$

$$y = Cx + Du.$$

The controllability matrix is :

$$J_c[A, B] = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

The model is completely controllable if and only if

$J_c[A, B]$ has full row rank.

Controller - Canonical Form:

For a completely controllable state space model:

$$A_c = \begin{bmatrix} -a_1 & -a_2 & \dots & \dots & -a_n \\ 1 & 0 & & & 0 \\ 0 & 1 & 0 & & 0 \\ \vdots & 0 & & \ddots & 0 \\ 0 & 0 & & & 1 & 0 \end{bmatrix} \quad B_c = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$C_c = [b_1 \ b_2 \ \dots \ b_n]$$

eg. $G(s) = \frac{5}{s^2 + 3s + 12}$

$$\Rightarrow A_c = \begin{bmatrix} -3 & -12 \\ 1 & 0 \end{bmatrix} \quad B_c = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & -12 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$